

SEISMIC BUMPS

HAZARD PREDICTION

Predicting high-energy seismic bumps
using data-driven classification models.

PRIYADHARSHINI P

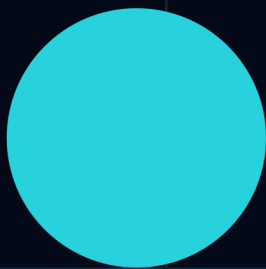
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SEISMIC ACTIVITY



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A clean project roadmap



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INTRODUCTION

Why seismic bump prediction matters

- Seismic activity is an important safety concern in underground coal mining.
- This project analyzes seismic activity data using Machine Learning.
- The main goal is to predict whether a high-energy seismic bump will occur in the next shift.
- The task is treated as a binary classification problem.
- The target variable is class: 1 = hazardous state, 0 = non-hazardous state.

PREDICTION FLOW

-  **01**
SEISMIC DATA
-  **02**
FEATURES
-  **03**
ML MODEL
-  **04**
CLASS

DATASET SOURCE & DETAILS

UCI Machine Learning Repository • Dataset ID 266

2,584

INSTANCES

19

ATTRIBUTES

Classification

8 HRS

PER SHIFT

DATASET PROFILE

- Source: UCI Machine Learning Repository
- Dataset: Seismic Bumps
- Collected from two longwalls in a Polish coal mine
- Each row represents one 8-hour shift
- Task: Binary classification

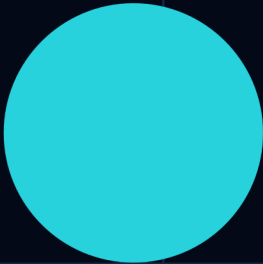
TARGET VARIABLE

class



Source: UCI Machine Learning Repository

DATASET DESCRIPTION



Key variables used for modeling



seismic
Seismic activity level



seismoacoustic
Seismoacoustic activity



genergy
Energy-related measure



gpuls
Pulse-related measure



nbumps
Number of seismic bumps



energy
Seismic energy

NMOTE

maxenergy
The dataset contains 19 attributes including the target variable and has an imbalanced class distribution.
Maximum seismic energy



EXPLORATORY DATA ANALYSIS

Understand patterns before training

EDA CHECKLIST

Statistical analysis

- a. df.head()
- b. df.tail()
- c. df.info()
- d. df.describe()
- e. df.shape
- f. df.columns

Handling missing values

- a. df.isnull().sum()

Handling duplicates

- a. df.duplicated().sum()

Finding Correlation

- a. c = df.corr() display using heatmap

Skewness

Encoding Conversion of categorical to numerical

- a. Label encoding - only target variable/y variable
- b. OneHotEncoding - for all independent variable /X variable

Handling Outliers

- display using Boxplot

SMOTE

Balance the target value

Feature Selection

Select the relevant feature with SelectKBest algorithm

Scaling

Standard Scaling

TARGET DISTRIBUTION

CLASS 0



CLASS 1



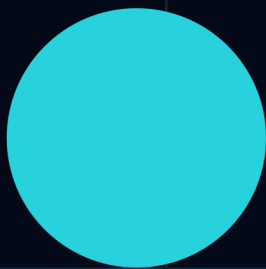
Highly imbalanced target → use suitable evaluation metrics

FEATURE INSIGHTS



DATA PREPROCESSING

Prepare clean and model-ready data



07



CHECK

Missing values & duplicates



ENCODE

Convert categorical features



SPLIT

Features X + target y



SCALE

Normalize where required



BALANCE

Handle class imbalance

GOAL → Convert raw seismic observations into reliable inputs for binary classification.



MACHINE LEARNING ALGORITHMS

Compare multiple classification approaches



LOGISTIC REGRESSION

Baseline linear classifier



DECISION TREE

Rule-based nonlinear model



RANDOM FOREST

Ensemble of decision trees



KNN

Distance-based classifier

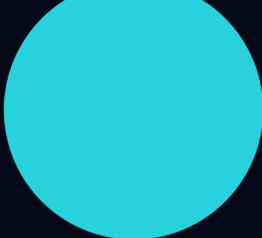


SVM

Margin-based classifier

TRAIN → PREDICT → EVALUATE

MODEL EVALUATION & COMPARISON



Metrics that matter for an imbalanced dataset

ACCURACY

0.53

Overall correctness

PRECISION

0.52

Correct positive predictions

RECALL

0.84

Positive cases detected

F1-SCORE

0.63

Balance of precision & recall

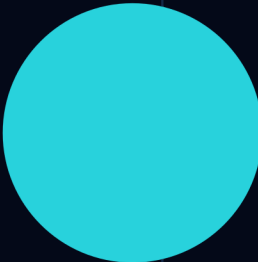
MODEL	ACCURACY	PRECISION	RECALL	F1-SCORE
Logistic Regression	0.48	0.49	0.59	0.53
Decision Tree	0.49	0.49	0.49	0.49
Random Forest	0.48	0.48	0.47	0.48
Gradient Boosting	0.50	0.50	0.49	0.50
KNN	0.51	0.51	0.50	0.51
SVM	0.53	0.52	0.65	0.58
Naive Bayes	0.51	0.50	0.84	0.63

MODEL COMPARISON



BEST MODEL & RESULTS

Replace the placeholders with your final run



BEST MODEL

Naive Bayes

Selected using the strongest overall evaluation performance.



ACCURACY

51 %

PRECISION

50 %

RECALL

84 %

F1-SCORE

63 %

Use your actual scores from the Python model comparison.



CONCLUSION

Key takeaways

- The Seismic Bumps dataset was analyzed using Machine Learning classification techniques.
- EDA and preprocessing helped prepare the seismic data for modeling.
- Multiple classification algorithms were trained and compared.
- The best model was selected using appropriate evaluation metrics.
- Machine Learning can support forecasting of increased seismic activity and safer mining operations.

PROJECT OUTCOME



Thank You